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Theme 3.1 Business Economics — Edexcel B

1. Business Growth

Business growth: An increase in the size of a firm, measured by output, revenue, market share, number of employees, or asset value. Firms grow to increase profits, exploit economies of scale, gain market power, and reduce risk through diversification.

1.1 Why Firms Grow

- **Economies of scale:** Larger firms can reduce average costs, improving competitiveness and profit margins.
- **Market power:** A larger market share gives more pricing power and bargaining power with suppliers.
- **Risk diversification:** Operating in multiple markets or products reduces dependence on any single one.
- **Managerial motives:** Managers may pursue growth to increase their own salaries, prestige, and power (Marris's model).
- **Shareholder pressure:** Investors may demand growth to increase share value and returns.
- **Survival:** In competitive markets, firms that do not grow may be outcompeted or acquired.

1.2 Methods of Growth

Internal (Organic) Growth

Growth from within the firm — expanding output, opening new stores, developing new products, entering new markets, investing in capacity.

- **Advantages:** Less risky, easier to manage, retains corporate culture, financed from retained profits.
- **Disadvantages:** Slower, may miss market opportunities, limited by internal resources.

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External (Inorganic) Growth — Mergers and Takeovers

Type	Description	Example	Motivation
Horizontal integration	Merger with a firm at the same stage of production in the same industry	Facebook acquiring Instagram	Economies of scale, increased market share, reduce competition
Vertical integration (forward)	Merger with a firm closer to the consumer	A brewery buying a pub chain	Control distribution, secure retail outlets, capture retail margins
Vertical integration (backward)	Merger with a firm closer to raw materials	A car maker buying a steel supplier	Control supply chain, secure raw materials, reduce costs
Conglomerate (lateral)	Merger with a firm in an unrelated industry	Virgin Group (airlines, telecoms, banking)	Diversify risk, spread into growth markets

Case Study: Amazon's Growth Strategy

Amazon exemplifies both organic and inorganic growth. Organic growth came from expanding product lines (from books to everything), launching Amazon Web Services (cloud computing, now its most profitable division), and building a massive logistics network. Inorganic growth included acquiring Whole Foods (2017, forward vertical integration into grocery retail), MGM Studios (2022, content for Prime Video), and Ring (smart home devices). Amazon's strategy demonstrates how firms combine growth methods to achieve market dominance across multiple industries, leveraging economies of scope and network effects.

1.3 Constraints on Growth

- **Diseconomies of scale:** As firms grow very large, average costs may rise due to coordination problems, communication breakdowns, and loss of motivation.
- **Finance:** Growth requires investment — small firms may struggle to access capital.

- **Regulation:** Competition authorities (CMA in the UK) may block mergers that reduce competition.
- **Market size:** In niche markets, there may be limited scope for growth.
- **Managerial capacity:** The firm may lack the management talent to oversee rapid expansion.

1.4 Demergers

> **Demerger:** The splitting of a firm into two or more separate businesses. This may occur when a division is underperforming, when a conglomerate becomes too complex to manage, or when a focused firm is valued more highly by investors than a diversified one.

Examples: eBay splitting from PayPal (2015); GSK splitting into GSK (pharmaceuticals) and Haleon (consumer health) in 2022.

2. Costs of Production

2.1 Short Run vs Long Run

> **Short run:** The time period in which at least one factor of production is fixed (typically capital — a firm cannot quickly build a new factory).
>
> **Long run:** The time period in which all factors of production are variable — the firm can change the scale of all its inputs.

2.2 Fixed, Variable and Total Costs

Cost Type	Definition	Examples
Fixed costs (FC)	Costs that do not vary with output in the short run	Rent, insurance, salaries of permanent staff, interest on loans
Variable costs (VC)	Costs that vary directly with output	Raw materials, energy, wages of temporary staff, packaging

Total cost (TC)	TC = FC + VC	All costs of production at a given output level

2.3 Average and Marginal Costs

$$\text{Average Total Cost (ATC)} = \text{TC} \div \text{Q} = \text{AFC} + \text{AVC}$$

$$\text{Average Fixed Cost (AFC)} = \text{FC} \div \text{Q}$$

$$\text{Average Variable Cost (AVC)} = \text{VC} \div \text{Q}$$

$$\text{Marginal Cost (MC)} = \text{Change in TC} \div \text{Change in Q}$$

Key relationships:

- AFC falls continuously as output rises (fixed costs spread over more units).
- ATC is U-shaped — initially falls as AFC is spread and efficiency improves, then rises as diminishing returns increase variable costs per unit.
- MC cuts ATC and AVC at their lowest points (the MC curve intersects the average curves at their minimum).
- When MC < ATC, ATC is falling. When MC > ATC, ATC is rising.

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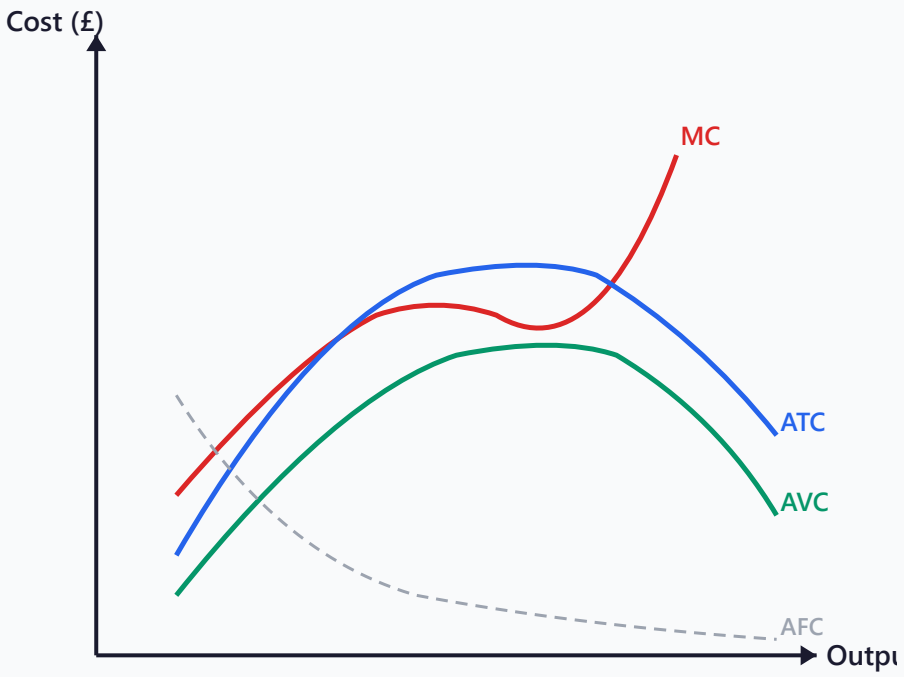

Figure 1: Short-run cost curves — MC crosses ATC and AVC at their minimum points. AFC falls continuously.

2.4 The Law of Diminishing Returns

Law of diminishing (marginal) returns: In the short run, as additional units of a variable factor (e.g., labour) are added to a fixed factor (e.g., capital), beyond a certain point, the marginal product (extra output from each additional worker) will begin to fall. This is a short-run concept only — it occurs because the fixed factor becomes over-utilised.

Diminishing returns explain why MC and AVC eventually rise — each extra worker adds less output, so the cost per extra unit increases.

3. Revenue

$$\text{Total Revenue (TR)} = \text{Price} \times \text{Quantity} = P \times Q$$

$$\text{Average Revenue (AR)} = \text{TR} \div \text{Q} = \text{Price (the demand curve)}$$

$$\text{Marginal Revenue (MR)} = \text{Change in TR} \div \text{Change in Q}$$

3.1 Revenue in Different Market Structures

- **Perfect competition:** The firm is a price taker — AR = MR = Price. Both are horizontal lines. TR rises linearly with output.
- **Imperfect competition (monopoly, oligopoly):** The firm faces a downward-sloping demand curve — AR falls as output rises. MR falls faster than AR (MR is below AR). TR rises, reaches a peak (where MR = 0), then falls.

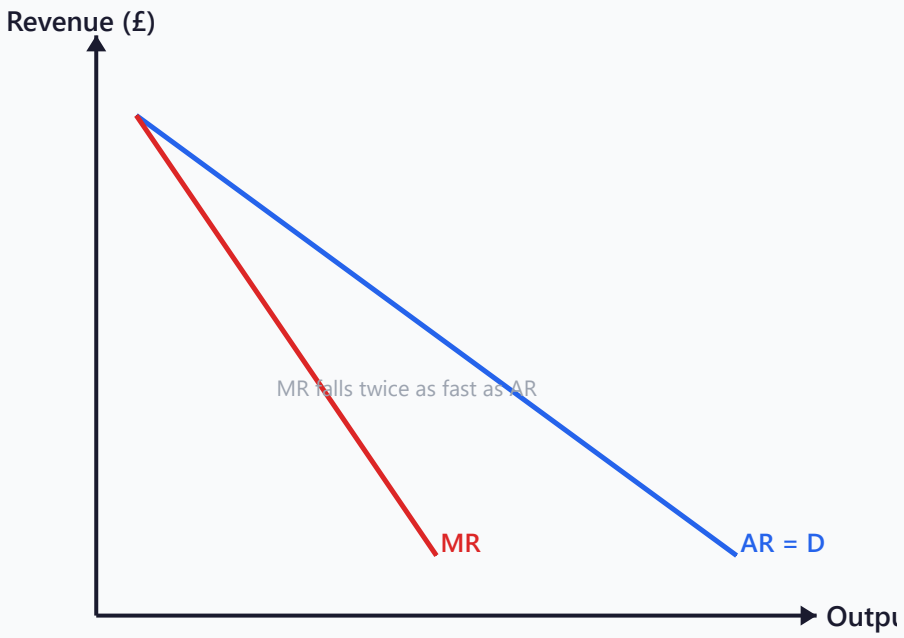

Figure 2: Revenue curves for a firm with market power — AR (demand) slopes down, MR lies below AR and falls at twice the rate (for a linear demand curve).

4. Economies and Diseconomies of Scale

Economies of scale: Reductions in long-run average cost (LRAC) as a firm increases its scale of production. They explain the downward-sloping section of the LRAC curve.

4.1 Internal Economies of Scale

Type	Explanation	Example
Technical	Large firms can use more efficient, specialised machinery and benefit from the **cube-square rule** (doubling dimensions of a container increases volume by 8x but surface area only 4x)	Large oil tankers have lower cost per barrel; car production lines
Purchasing (bulk buying)	Large firms can negotiate lower prices from suppliers due to bulk orders	Tesco negotiating lower wholesale prices than independent shops
Managerial	Large firms can employ specialist managers (HR, finance, marketing) rather than generalists	Multinationals employing dedicated tax specialists
Financial	Large firms can borrow at lower interest rates (seen as lower risk) and access a wider range of finance (bond markets, share issues)	FTSE 100 firms borrowing at lower rates than SMEs
Marketing	Advertising costs are spread over a larger output — lower cost per unit sold	National TV advertising for Coca-Cola costs fractions of a penny per bottle
Risk-bearing	Large firms can diversify across products and markets, reducing risk	Unilever operating in 190 countries across food, cleaning products, and personal care

4.2 External Economies of Scale

Cost reductions that benefit all firms in an industry as the industry itself grows:

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- **Skilled labour pool:** A concentration of firms attracts specialist workers (e.g., tech workers in Silicon Valley, financial services workers in the City of London).
- **Infrastructure:** Transport links, broadband, and other infrastructure develop around industrial clusters.
- **Knowledge spillovers:** Ideas and innovations spread between firms in a cluster.
- **Specialist suppliers:** Component suppliers and support services locate near clusters of firms.

4.3 Diseconomies of Scale

Diseconomies of scale: Increases in long-run average cost as a firm grows beyond its optimal size. They explain the upward-sloping section of the LRAC curve.

- **Communication problems:** In very large organisations, messages may be distorted as they pass through many layers of hierarchy.
- **Coordination difficulties:** Managing multiple divisions across countries becomes complex and costly.
- **Loss of motivation:** Workers in large firms may feel like a small cog in a big machine, reducing effort and productivity.
- **Principal-agent problem:** Owners (principals) cannot effectively monitor managers (agents) in very large firms.

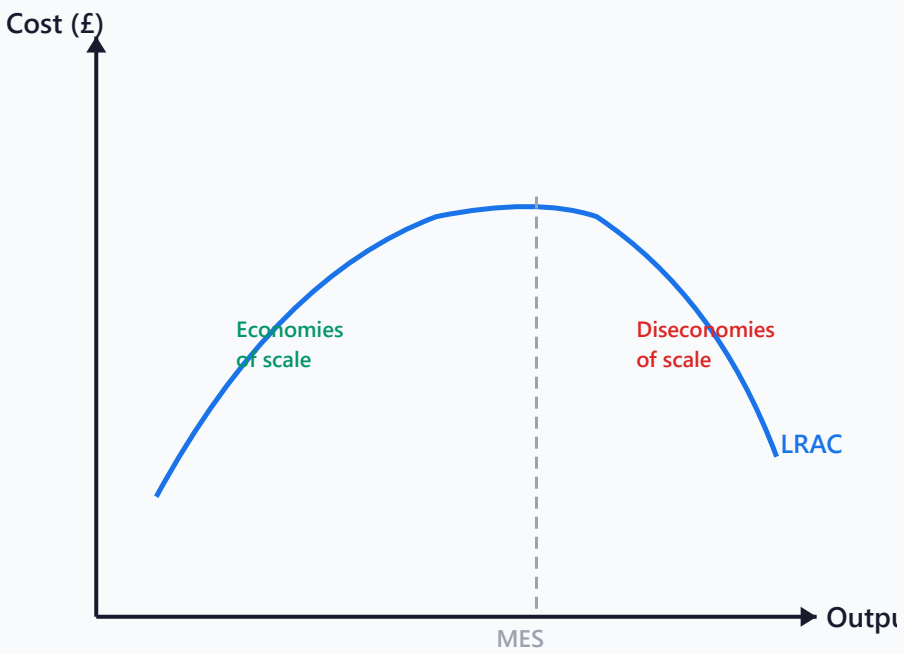

Figure 3: The long-run average cost (LRAC) curve. MES = minimum efficient scale — the lowest output at which LRAC is minimised.

4.4 Minimum Efficient Scale (MES)

MES: The minimum level of output at which long-run average costs are minimised. Industries with a high MES relative to market size tend to have fewer, larger firms (natural tendency towards oligopoly or monopoly). Industries with a low MES can support many small firms.

Case Study: Economies of Scale in the UK Supermarket Industry

The UK grocery market is dominated by four large chains — Tesco (27% market share), Sainsbury's (15%), Asda (14%), and Morrisons (9%). These firms exploit enormous economies of scale: purchasing economies (bulk buying from farmers and suppliers at lower per-unit costs), technical economies (automated distribution centres and self-checkout technology), marketing economies (national advertising campaigns spread over billions of products), and financial economies (access to cheap bank lending and bond markets). The MES in grocery retail is very high, which is why smaller independent shops struggle to compete on price. However, discounters Aldi and Lidl have grown rapidly by focusing on a narrow product range with extremely lean operations.

5. Profit

$$\text{Total Profit} = \text{Total Revenue} - \text{Total Cost}$$

$$\text{Profit per unit} = \text{AR} - \text{ATC}$$

5.1 Normal Profit vs Supernormal Profit

Normal profit: The minimum level of profit needed to keep a firm in the industry in the long run. It is the opportunity cost of enterprise — the return the entrepreneur could earn in the next best alternative use of their time and capital. Normal profit occurs where TR = TC (including the

entrepreneur's opportunity cost in total cost).

Supernormal (abnormal/economic) profit: Profit above normal profit — TR exceeds TC. This is the profit that attracts new firms into an industry (in competitive markets) or is sustained behind barriers to entry (in monopoly/oligopoly).

5.2 Profit Maximisation

A firm maximises profit where **MC = MR** (marginal cost equals marginal revenue), provided MC is rising through MR. At this point, the revenue from the last unit exactly equals its cost — producing any more would add more to cost than revenue.

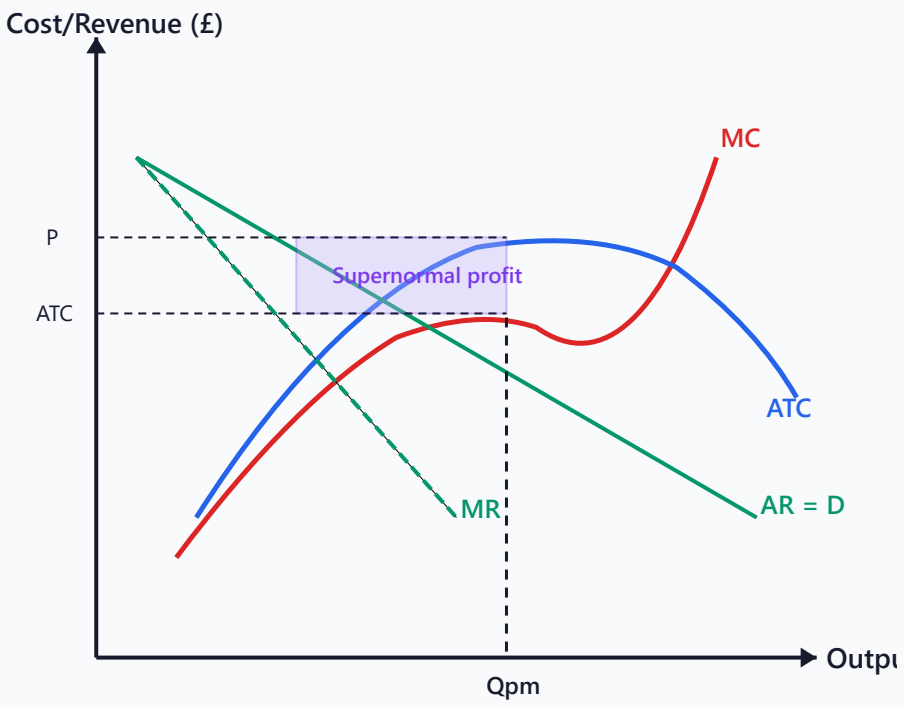

Figure 4: Profit maximisation at MC = MR. Supernormal profit shown as the shaded rectangle between price (AR) and ATC at Qpm.

5.3 The Shut-Down Condition

- **Short run:** A firm should continue producing as long as price (AR) covers average variable cost (AVC). If AR < AVC, the firm should shut down — it is not even covering its variable costs.

- **Long run:** A firm should exit the industry if AR < ATC — it is not making normal profit.

6. Business Objectives

Not all firms pursue profit maximisation. Alternative objectives may reflect different stakeholder priorities:

Objective	Rule	Explanation
Profit maximisation	MC = MR	Traditional assumption — maximise the difference between TR and TC. Most relevant for owner-managed firms.
Revenue maximisation	MR = 0	Managers may prefer to maximise revenue (and their own bonuses linked to sales) rather than profit (Baumol's model). Output is higher than at profit max.
Sales maximisation	AR = ATC (normal profit constraint)	Firm sells as much as possible while still covering costs (Baumol). May be used to build market share or deter entry.
Satisficing	Earn "enough" profit to keep stakeholders happy	Herbert Simon — firms face bounded rationality and conflicting stakeholder demands. They aim for a satisfactory level of profit rather than the maximum (realistic for large firms with complex governance).
Survival	Cover costs, avoid losses	In recessions or for start-ups, simply staying in business may be the primary goal.
Social/ethical objectives	Varies — consider wider stakeholders	Social enterprises, cooperatives, or firms with strong CSR commitments (e.g., Patagonia, The Body Shop).

6.1 The Principal-Agent Problem

Principal-agent problem: A conflict of interest that arises when the owners (principals/shareholders) of a firm have different objectives from the managers (agents) who run it

day-to-day. Shareholders typically want profit maximisation, but managers may pursue growth, revenue maximisation, or perks (empire building). This is particularly acute in large public limited companies where ownership is dispersed.

Solutions: Performance-related pay, share option schemes (align manager and shareholder interests), non-executive directors, threat of takeover (disciplining management), improved corporate governance.

Case Study: Short-Termism in UK PLCs

UK stock market-listed companies have been criticised for short-termism — prioritising quarterly earnings and share price over long-term investment. The Kay Review (2012) found that average shareholding periods in UK companies had fallen from about 5 years in the 1960s to less than 7 months. This pressure from shareholders encourages managers to cut costs (including R&D and training) to boost short-term profits rather than investing in long-term productivity growth. The consequence is lower long-term competitiveness — contributing to the UK's persistent productivity gap compared to France, Germany, and the US.

Case Study: John Lewis Partnership

John Lewis operates as a partnership owned by its employees ("Partners"), illustrating an alternative business objective. Rather than maximising shareholder profit, the Partnership aims to satisfy all stakeholders — employees receive a share of annual profits as a bonus, and the firm emphasises long-term sustainability, employee welfare, and customer service. This model challenges the assumption that profit maximisation is the universal objective of firms, though John Lewis has faced financial difficulties in recent years, raising questions about whether this model can compete effectively with profit-driven rivals.

Evaluation: Business Objectives

- In practice, most firms pursue multiple objectives simultaneously, with the priority shifting based on market conditions.
- In highly competitive markets, firms are effectively forced to profit maximise — those that do not will be driven out by more efficient rivals.
- In monopoly or oligopoly, firms have more discretion to pursue non-profit objectives because supernormal profits provide a buffer.

- The principal-agent problem means that in large PLCs, managers often have more influence over objectives than shareholders.
- Short-run and long-run objectives may differ — a firm may sacrifice short-run profit to build market share (e.g., Amazon operated at a loss for years).

Exam Tip: When analysing a firm's behaviour, always consider: (1) What is its most likely objective? (2) Is it in a competitive or concentrated market? (3) Is it owner-managed or shareholder-owned? (4) What time horizon is relevant? This context determines which model of firm behaviour to apply.

7. Evaluating Business Growth and Mergers

Evaluation: Does Merger Growth Create Value?

- **Evidence against:** Academic studies (e.g., McKinsey, KPMG) consistently find that 50-70% of mergers fail to create shareholder value. Acquirers frequently overpay ("winner's curse"), overestimate synergies, and underestimate cultural integration difficulties.
- **Managerial motives:** Marris's growth maximisation model argues managers pursue growth to increase their own power, prestige, and remuneration — even when this destroys shareholder value. Empire-building is a manifestation of the principal-agent problem.
- **Horizontal mergers and market power:** Competition authorities (CMA in UK, European Commission, US FTC/DoJ) may block mergers that substantially reduce competition. The CMA blocked the Microsoft/Activision deal in 2023 — later modified and approved — and blocked the Meta/Giphy acquisition. This constrains how large firms can grow through horizontal integration.
- **However:** Some mergers do create genuine value through economies of scale, knowledge transfer, and access to new markets. Amazon's acquisition of Whole Foods gave it immediate physical retail infrastructure and data on grocery shopping — hard to replicate organically.
- **Organic vs inorganic:** Organic growth is slower but less risky and preserves culture. Inorganic growth is faster but involves integration challenges. The "best" method depends on the firm's resources, the competitive environment, and management capability.

8. Evaluation: Economies of Scale and Market Structure

Are Economies of Scale Always Beneficial?

- **Consumer prices:** If passed on, economies of scale reduce costs and prices — benefiting consumers (e.g., Aldi/Lidl's lean operating model delivers low prices). However, if not passed on, the cost savings become supernormal profit for the firm.
- **Market concentration:** High MES relative to market size tends toward monopoly or oligopoly. This may deliver productive efficiency (low unit costs) but at the expense of allocative efficiency (prices above MC, restricted output). The regulator must balance these.
- **Barriers to entry:** High fixed costs (a source of technical economies of scale) act as a barrier to entry — protecting incumbent firms from competition. This can entrench monopoly power and reduce innovation over time.
- **Diseconomies in practice:** Royal Bank of Scotland before 2008 grew to become one of the world's largest banks by assets — then required a £45.5 billion government bailout due to organisational complexity, poor communication across divisions, and excessive risk-taking. Its collapse illustrates how diseconomies of scale (coordination failure, principal-agent problems) can be catastrophic.
- **X-inefficiency (Leibenstein):** Large firms sheltered from competition may become inefficient — not because average costs rise, but because the firm does not minimise costs. Privatisation and deregulation are often justified on these grounds.

9. Evaluation: Profit Maximisation as the Dominant Assumption

How Realistic Is the Profit Maximisation Assumption?

- **Owner-managed firms (SMEs):** Where the owner IS the manager, there is no principal-agent problem and profit maximisation is a reasonable assumption. Approximately 99% of UK businesses are SMEs — so the assumption applies broadly to most firms by number, if not by output.
- **Large PLCs:** The separation of ownership (shareholders) from control (managers) means managers may satisfice or pursue growth rather than profit. Evidence: UK CEO pay has grown far

faster than shareholder returns, suggesting managers capture value rather than create it.

- **Market structure matters:** In perfectly competitive markets, firms that do not produce at minimum LRAC are driven out by competition — so profit maximisation (or at least cost minimisation) is effectively forced. In monopoly, there is "slack" — firms can survive while satisficing.
- **Behavioural economics challenge:** Real decision-making is subject to cognitive biases, bounded rationality (Simon), and heuristics — firms do not always have the information or cognitive capacity to maximise anything precisely. "Rules of thumb" (e.g., cost-plus pricing) are common in practice.
- **The long-run argument:** Amazon explicitly sacrificed short-run profits for decades to build market share and infrastructure — consistent with a long-run profit maximisation strategy. Short-run profit maximisation is not the same as long-run profit maximisation.

Case Study: Tesla — Growth, Losses, and Long-Run Value

Tesla was founded in 2003 and went public in 2010, but did not report its first annual profit until 2020. During its first decade of losses, Tesla invested relentlessly in R&D, factory capacity, battery technology, and software. Revenue grew rapidly but expenses outpaced them. By the conventional short-run profit maximisation model, Tesla was failing. Yet investors valued the company at over $1 trillion at peak (2021), far ahead of traditional automakers, because the market believed Tesla's strategy would generate enormous future profits through market dominance in electric vehicles. Tesla illustrates that investors can distinguish between value-destroying losses (poor management) and value-creating losses (strategic investment) — and that short-run and long-run profit maximisation may be very different objectives in practice.

Exam Tips — Business Economics

- **Link theory to context:** Always state which market structure the firm is in before applying revenue/cost analysis. A perfectly competitive firm's AR = MR = P; a monopolist's MR < AR.
- **Use diagrams correctly:** Profit max is always at MC = MR (not MC = AR or MR = 0). Show the profit rectangle as (P - ATC) × Q, clearly labelling P and ATC on the y-axis.
- **Evaluate the objective:** Examiners reward discussion of whether profit maximisation is realistic. Mention the principal-agent problem, behavioural economics, and market structure as factors

determining which objective is most plausible.

- **Economies of scale diagram:** The LRAC is a U-shaped envelope of SRAC curves. Label MES clearly. Link MES to market structure — high MES → fewer firms → oligopoly/monopoly tendency.
- **Shut-down condition:** In the short run, produce if AR ≥ AVC (covering variable costs). In the long run, produce if AR ≥ ATC (covering all costs and earning normal profit). This is a frequently examined distinction.

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